Lecture 16:
Explainability

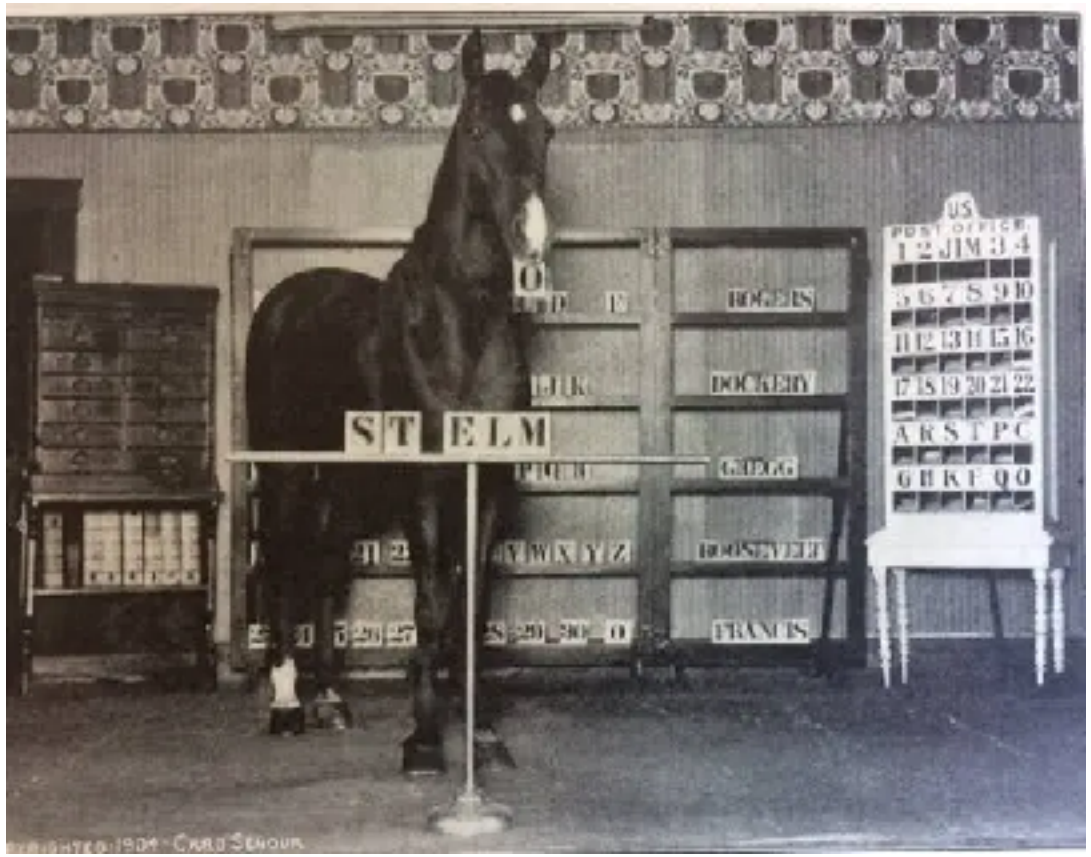
October 19, 2026

recap: so far we've accessed our neural networks on their outcome on test sets

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what is the model doing **inside**? Is it right for the right reason?

Clever Hans: right answers, wrong reasons



the German “calculating horse” in the 1900s "did arithmetic" by tapping his hoof

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was responding to the body language of his trainer

Lapuschkin et al (2019): image classifier that can classify horses exceptionally well



https://arxiv.org/pdf/1902.10178

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learned the copyright watermark on many horse photos … clever hans

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interpretable vs **explainable**

a model is **interpretable** if a human can follow how it maps inputs to outputs: a linear model, a shallow tree (*interpretability is built in)*

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we as astronomers care about explanations for *debugging*, *trust*, and *discovery*

interpretable by design

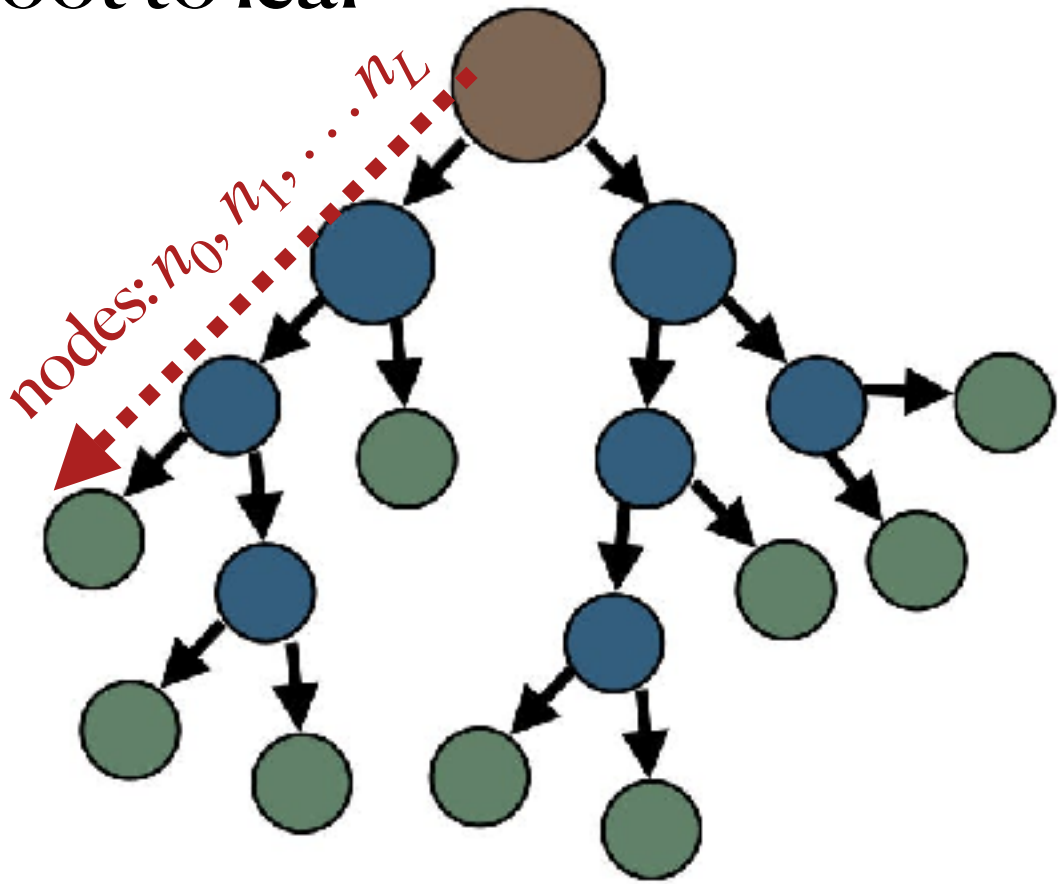
linear models: read the coefficients

$$f(x) = \beta_0 + \sum_j \beta_j x_j$$

β_j : change in the prediction per unit of x_j, others held fixed

decision trees: follow the path from root to leaf

$$f(x) = \bar{y}_{n_0} + \sum_{\ell=1}^{L} (\bar{y}_{n_\ell} - \bar{y}_{n_\ell - 1})$$

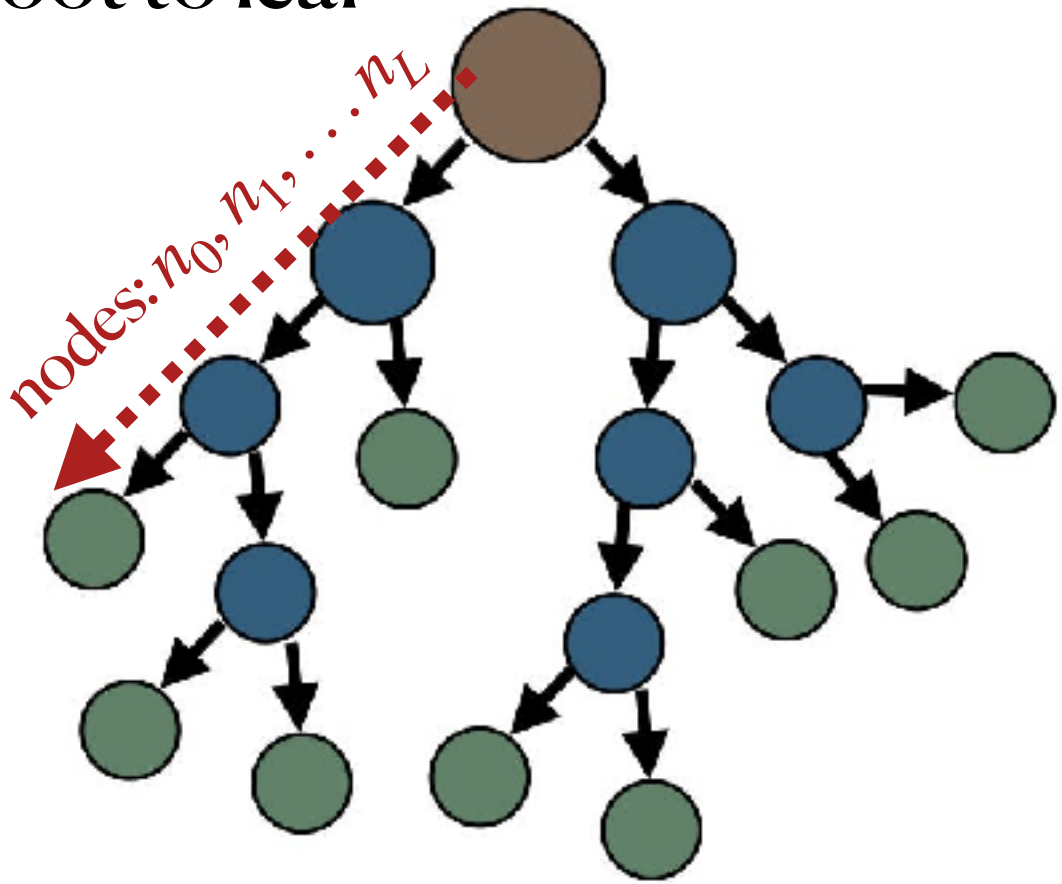


a regression tree predicts the mean of y in a leaf where each split changes the node mean

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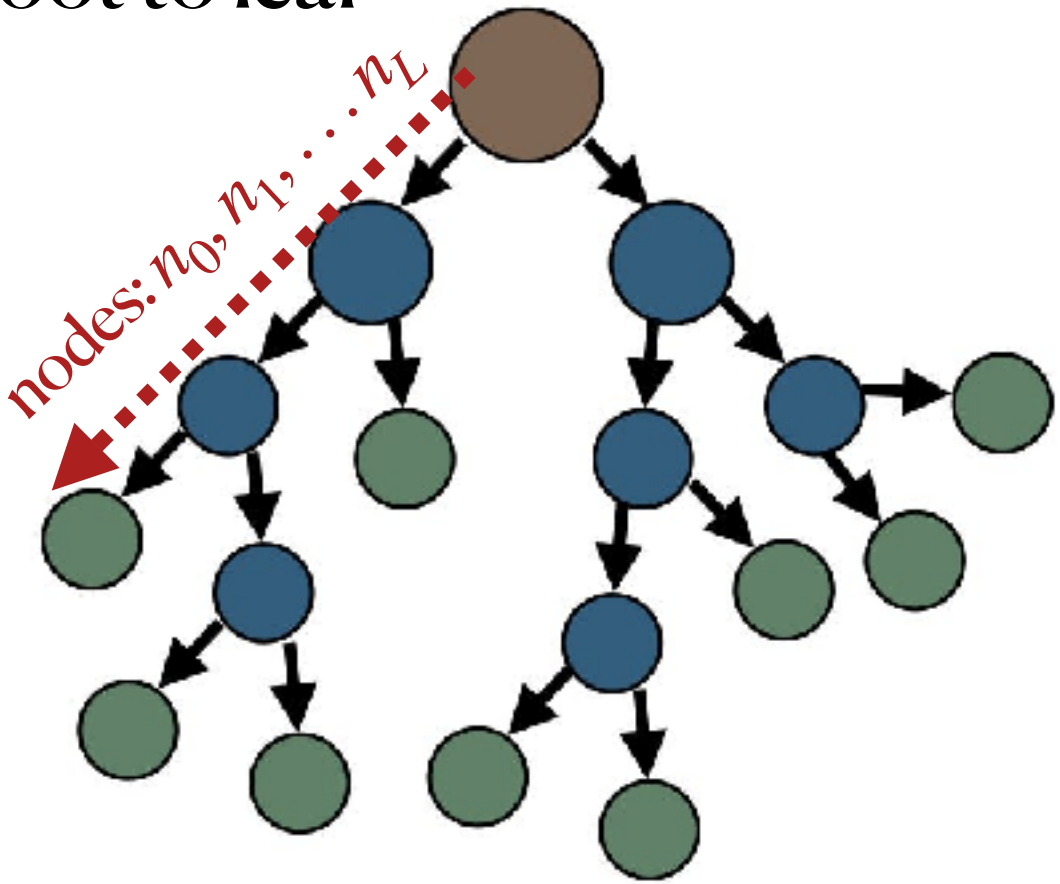
$$\text{contrib}_j(x) = \sum_{j} (\bar{y}_{n_\ell} - \bar{y}_{n_{\ell-1}})$$



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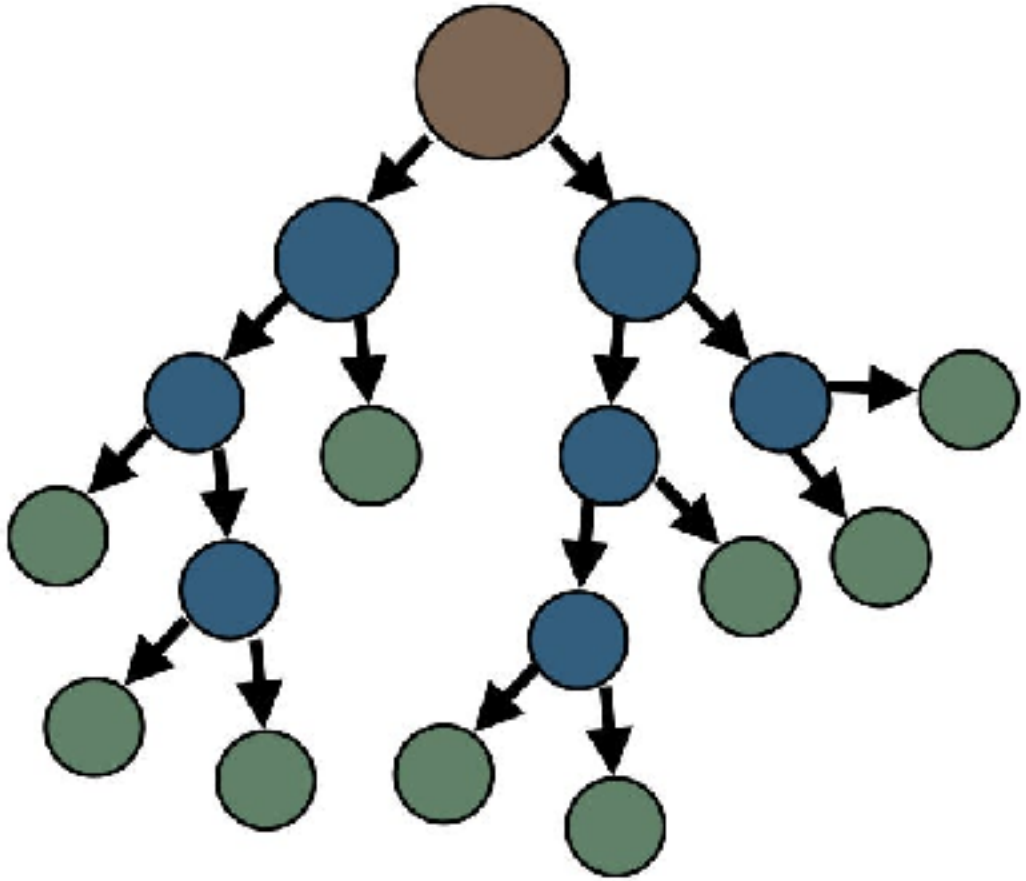


local (another x follows another path), if two features are correlated credit goes to whichever happens first

global importance

decision trees: Gini importance

recall Gini index $\sum_k p_k(1 - p_k)$



Every split lowers the Gini index (*impurity*)

Sum the weighted decrease over all splits on feature j

permutation importance: model agnostic, global importance

1. score the model on test data: e_{orig} (*e.g. MSE accuracy*)
2. shuffle data column j of the test set to break the link between x_j and y
3. rescore $e_{\text{perm},j}$ and calculate importance $e_{\text{perm},j}/e_{\text{orig}}$

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how much does the model rely on x_j

a shuffled x_j paired with the unshuffled, correlated x_k can make combinations that never occur then the model is scored on inputs it never saw.

leave one feature out: retrain model without feature j

$$e_{-j} = \text{score of model without feature } j$$

$$\text{then compare } e_{-j} / e_{\text{orig}}$$

is the information in x_j available elsewhere?

explaining one prediction

local interpretable model-agnostic explanations (**LIME**): *Ribeiro, Singh & Guestrin (2016)*

fit an interpretable surrogate model g to model f in a "neighborhood" of $\mathbf{x}$

$$g^\star = \arg\min_g \mathscr{L}(f, g, \pi_{\mathbf{x}}) + \Omega(g)$$

$$\mathscr{L} = \sum_k \pi_{\mathbf{x}}(\mathbf{z}_k)\Big(f(\mathbf{z}_k) - g(\mathbf{z}_k)\Big)^2$$

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penalty for complexity

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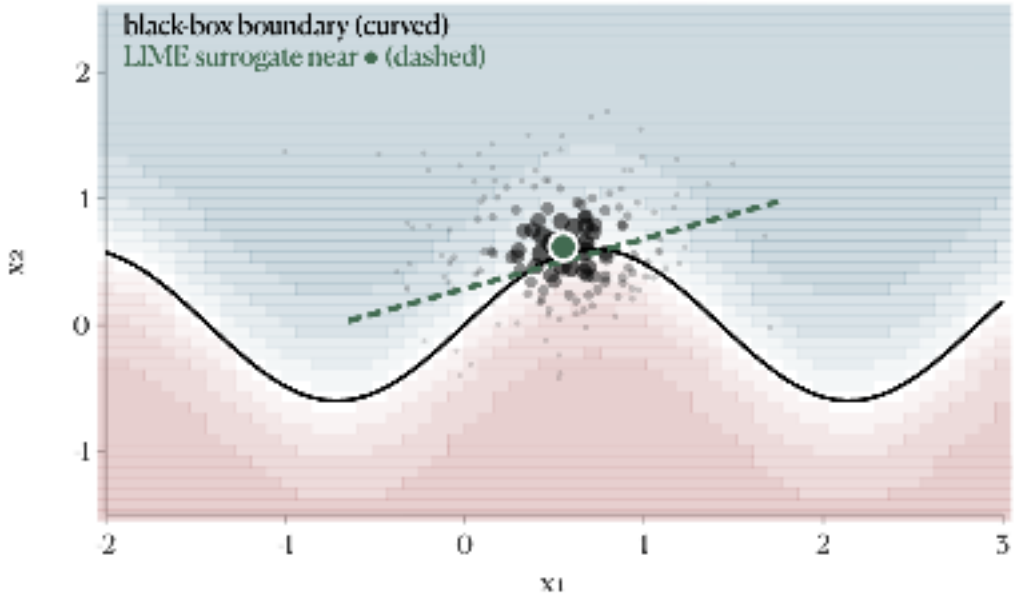
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***e.g.* linear surrogate model with L2 regularization**

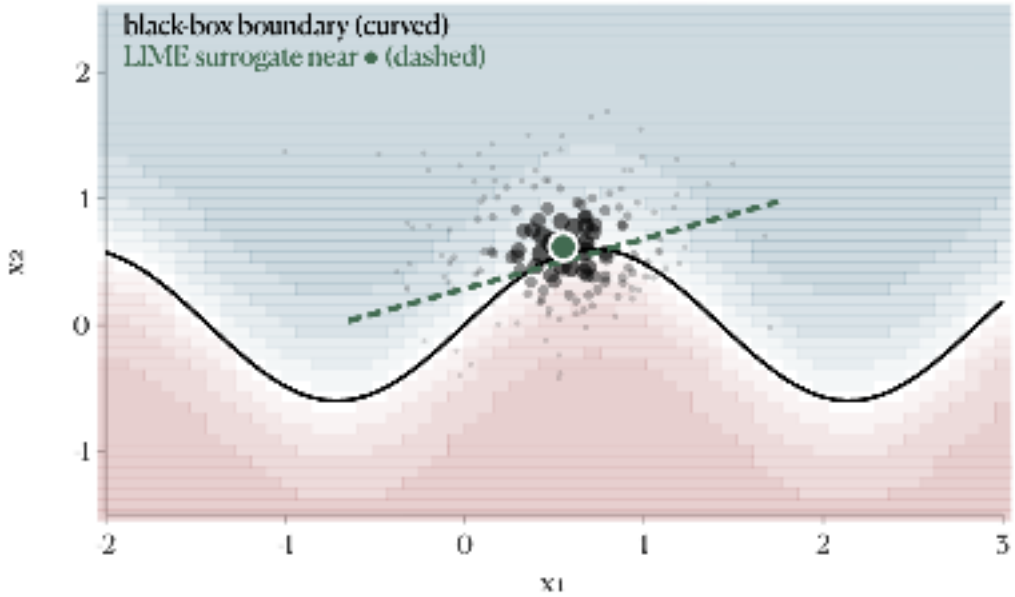
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LIME requires arbitrary choices (*kernel width, sampling, surrogate class*)

features as players in a game: cooperative game theory (*Shapley 1953*)

players form coalition S, $v(S)$ is the payout. *How do you split the payout fairly?*

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players: the p features of $\mathbf{x}$, $S \subseteq P$

payout $v(S)$: the expected prediction when the features in S are fixed to $\mathbf{x}$'s values and the rest are unknown

$$v(S) = \mathbb{E}_{\mathbf{X}_{\bar{S}}}\left[f(\mathbf{x}_S, \mathbf{X}_{\bar{S}})\right]$$

$$v(\varnothing) = \mathbb{E}\left[f(\mathbf{X})\right] \text{ and } v(P) = f(\mathbf{x})$$

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$$\text{Shapley value of feature } j\text{: } \phi_j = \sum_{S \subseteq P \setminus \{j\}} \frac{|S|!(p - |S| - 1)!}{p!} \left[v(S \cup \{j\}) - v(S) \right]$$

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$\phi_j = j$**'s average marginal contribution over all orders**

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$$\textit{e.g. } f(\mathbf{x}) = 2x_1 + x_2 + 3x_1x_3$$

Independent features with mean 0; explain $x = (1,2,1)$, where $f(x) = 7$ and $\mathbb{E}[f] = 0$

S	v(S)	S	v(S)
∅	0	{1,2}	4
{1}	2	{1,3}	5
{2}	2	{2,3}	2
{3}	0	{1,2,3}	7

$$\textit{e.g. } v(\{1\}) = 2 + \mathbb{E}[X_2] + 3 \times \mathbb{E}[X_3] = 2$$

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$$\phi_1 = \tfrac{1}{3}(2-0)+\tfrac{1}{6}(4-2)+\tfrac{1}{6}(5-0)+\tfrac{1}{3}(7-2) = 3.5,$$
$$\phi_2 = \tfrac{1}{3}(2-0)+\tfrac{1}{6}(4-2)+\tfrac{1}{6}(2-0)+\tfrac{1}{3}(7-5) = 2,$$
$$\phi_3 = \tfrac{1}{3}(0-0)+\tfrac{1}{6}(5-2)+\tfrac{1}{6}(2-2)+\tfrac{1}{3}(7-4) = 1.5,$$

$\phi_j = j$**'s average marginal contribution over all orders**

Shapley values in practice: SHAP, KernelSHAP, TreeSHAP...

SHAP importance: $\overline{|\phi_j|}$, mean absolute Shapley value over many **x**

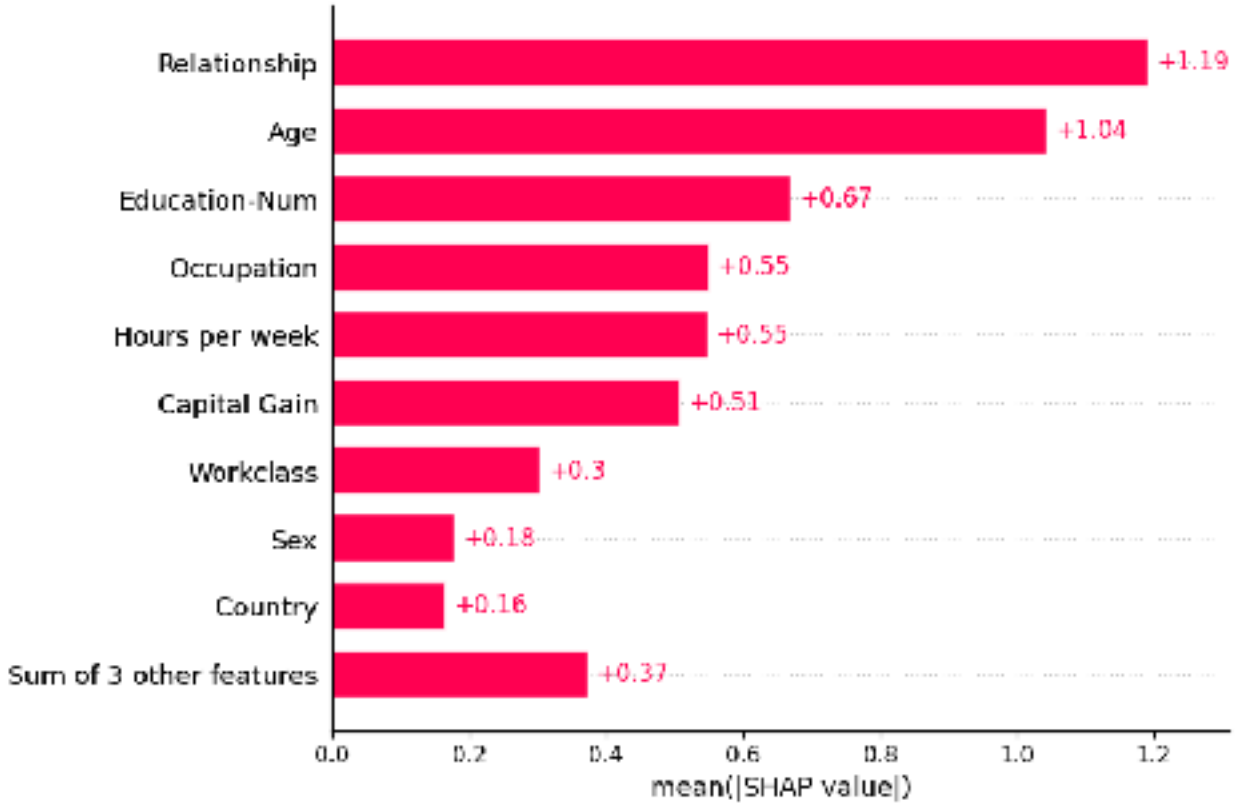


image: SHAP documentation

saliency maps

galaxy images have $34 \times 34 \times 3 = 3468$ pixels, Shapley values are out of reach but which pixels matter?

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how does the score change for an infinitesimal change of pixel i

Simonyan, Vedaldi & Zisserman (2014)

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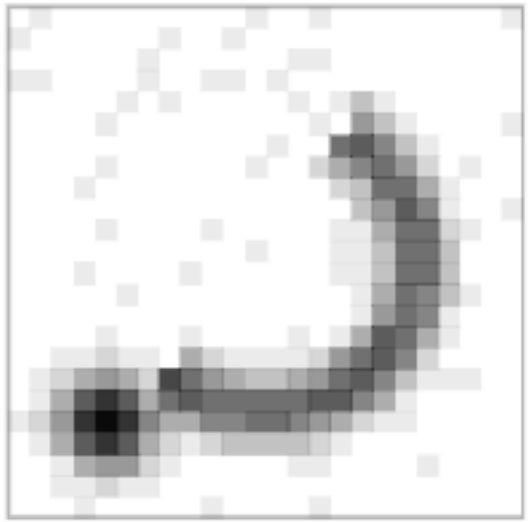
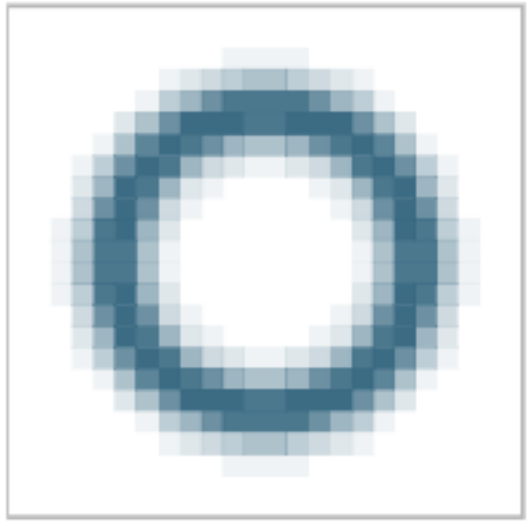


image x



gradient (×391 to show)

a "ring detector" $f(\mathbf{x}) = 4 \tanh(\langle \mathbf{w}, \mathbf{x} \rangle / \tau)$

where $\mathbf{w}$ is a ring template and $\tau = 0.15$

$$\langle \mathbf{w}, \mathbf{x} \rangle / \tau = 3.7 \text{ and } f = 3.99$$

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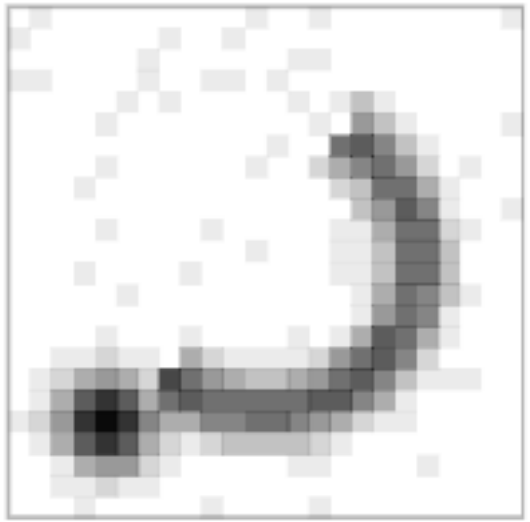
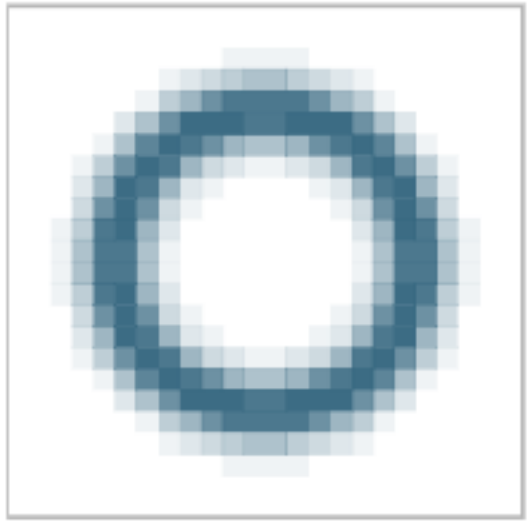


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$\nabla f = 0.0026$ and shows the whole template

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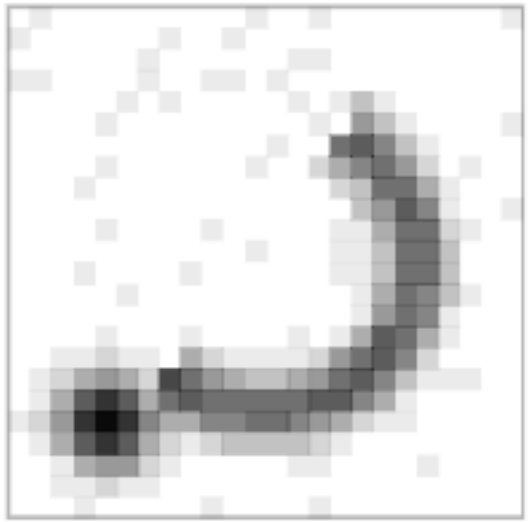
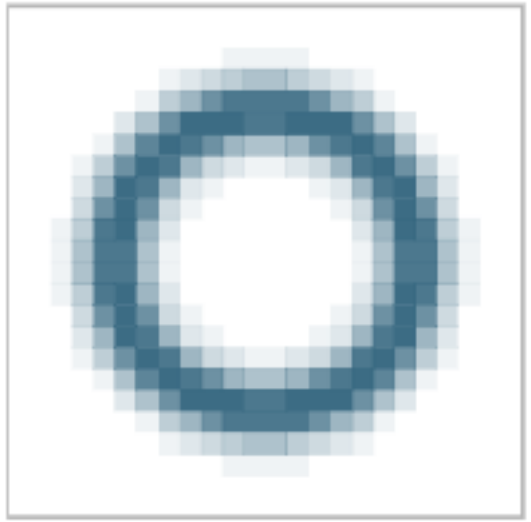


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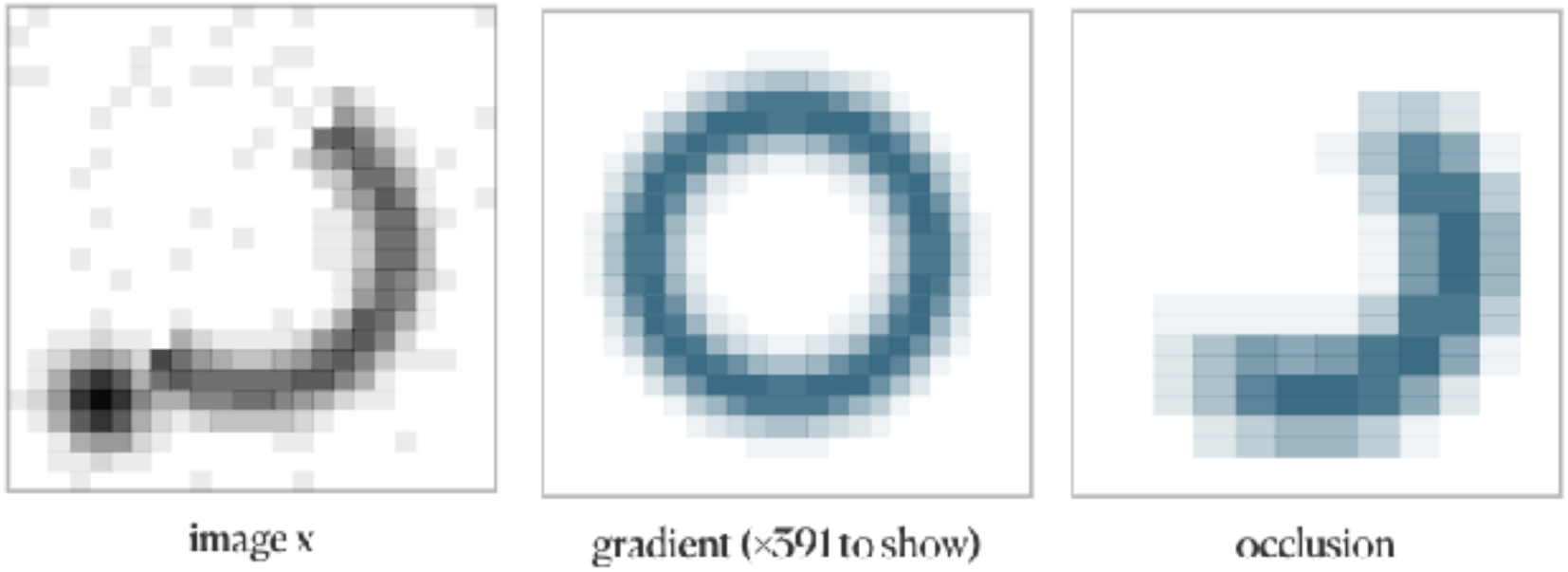
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describes *sensitivity* not contribution

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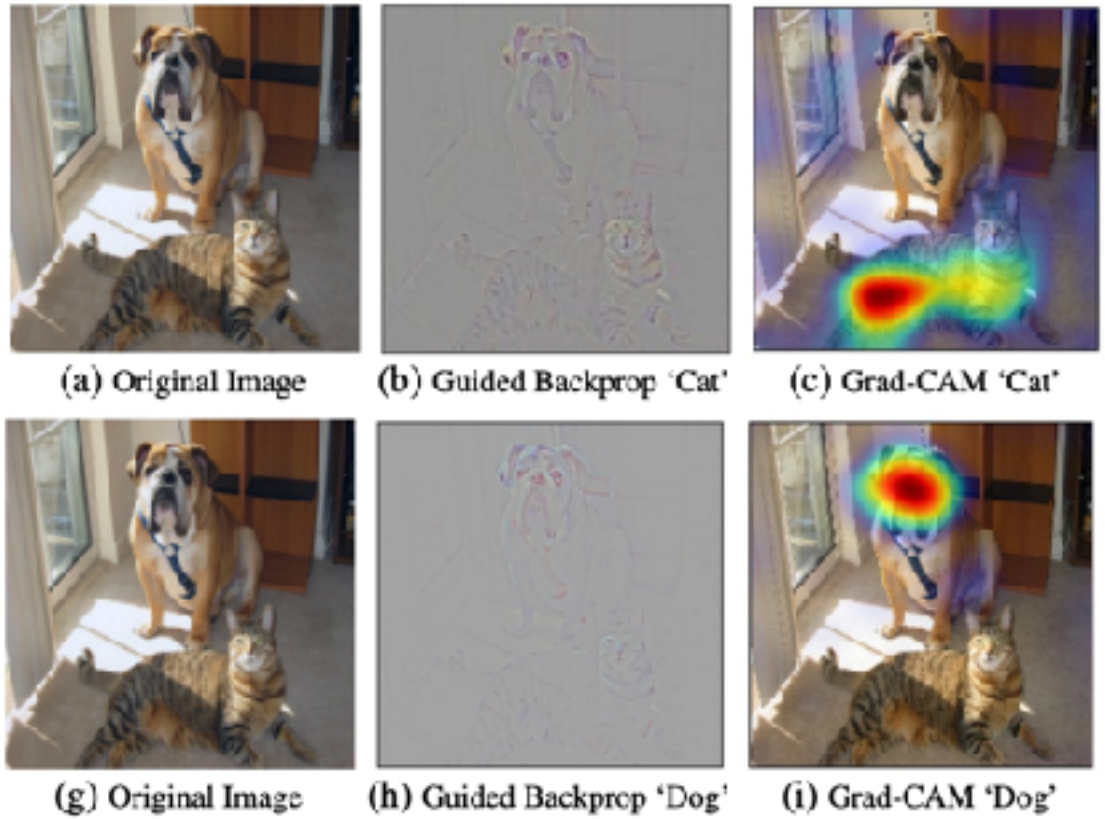
hide part of the image and see how much the prediction drops



Zeiler & Fergus (2014)

galaxy images have $34 \times 34 \times 3 = 3468$ pixels, Shapley values are out of reach but which pixels matter? **Grad-CAM**

last convolutional layer with K feature maps A^k (spatial, high level)



Selvaraju et al. (2017)

in-class: explain the photo-z MLP and CNN galaxy classifier

ast384c/inclass/lec16_interp/explain.ipynb

the point of these assignments is **not** the finished answer. It's what happens in your head while you produce it. *Don't let the LLM do the thinking*.

before using an LLM, ask yourself: am I trying to *understand* something, or trying to *avoid* understanding it? The first is resourceful. The second is outsourcing."

use it as a tutor, not a ghostwriter